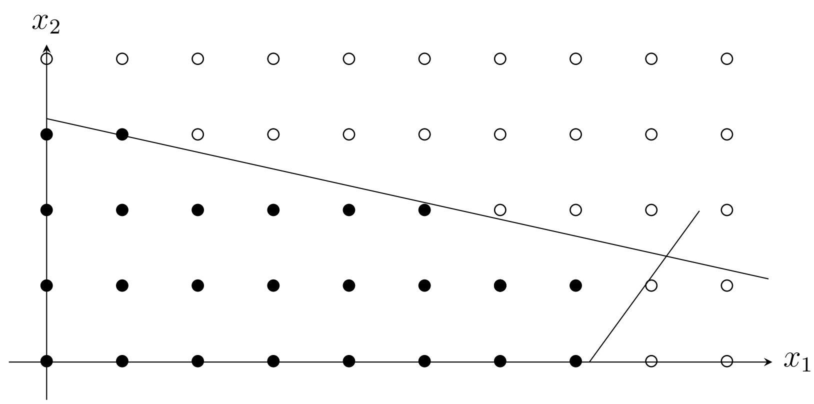
# Gomory cuts

Michel Bierlaire

*Solution of the practice quiz*

The feasible region is shown on the Figure below, where the plain dots are feasible, and the empty dots are outside of the polyhedron.




After the introduction of the slack variables, we obtain the following relaxation.

$$\min_x -3x_1 - 13x_2$$

subject to

$$\begin{aligned} 2x_1 + 9x_2 + x_3 &= 29, \\ 11x_1 - 8x_2 + x_4 &= 79, \\ x_1, x_2, x_3, x_4 &\geq 0. \end{aligned}$$

The optimal tableau of the relaxation is

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | | |
|---|---|---|---|---|---|
| 0 | 1 | 0.10 | -0.02 | 1.4 | $x_2$ |
| 1 | 0 | 0.07 | 0.08 | 8.2 | $x_1$ |
| 0 | 0 | 1.45 | 0.01 | 42.8 | |

The two variables are fractional, so two Gomory cuts can be generated. The first cut is

$$x_2 + \lfloor 0.1 \rfloor x_3 \lfloor -0.02 \rfloor x_4 \leq \lfloor 1.4 \rfloor,$$

that is

$$x_2 - x_4 \leq 1.$$

The second cut is

$$x1 + \lfloor 0.07 \rfloor x_3 + \lfloor 0.08 \rfloor x_4 \leq \lfloor 8.2 \rfloor,$$

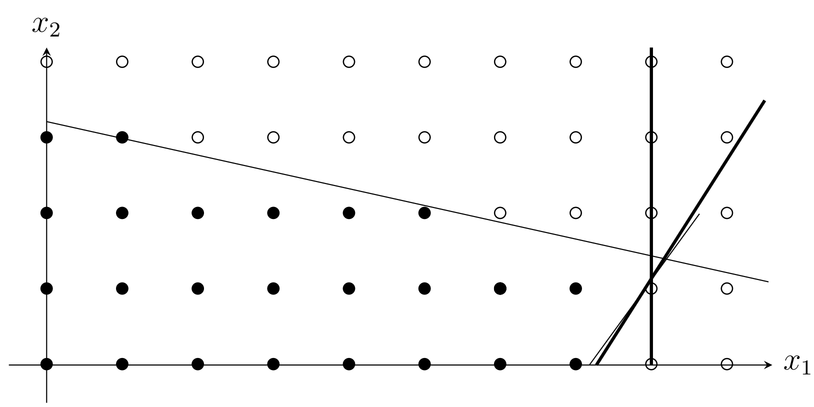
that is,

$$x_1 \leq 8.$$

As $x_4 = 79 - 11x_1 + 8x_2$, the first inequality in the original variables is written as

$$11x_1 - 7x_2 \leq 80.$$

These two cuts are illustrated in the Figure below:




It appears in this example that the part of the polyhedron that is cut is quite small. We expect, in this case, the algorithm to take a while to converge. Depending on the cut selected for inclusion at each iteration, the algorithm may use more than 50 iterations to find the optimal solution of this problem.

In practice, the cutting plane method is usually combined with the branch and bound algorithm. In addition of the additional constraints generated by the branching strategy, valid inequalities such as Gomory cuts are also included. Such a method is called *branch and cut*.